

# The Information Propagation Method for Efficient Network Reliability Analysis

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**Abstract**—In this paper, we present an ordered message-passing algorithm for propagating information related to system component conditions and dependency between nodes in the target graph network. This tool, called Information Propagation Method, exploits the structure of Directed Acyclic Graphs to assess the reliability of a network. The key advantage of this analytical approach is the computation efficiency obtained by maintaining a priority set of nodes to be updated in each step, every node in the graph is processed as infrequently as possible. Whilst the algorithm converges to an exact solution in simpler graph structures, the solution in more complex graphs is approximated due to the presence of the so-called *diamond subgraphs*. Diamond subgraphs are identifying solved and recursively decomposed by the algorithm allowing the exploration of complex component dynamics as well as system redundancy strategies. The algorithm is verified against the solution obtained via Monte Carlo simulation. Finally, we discuss a case study example showing the applicability and advantages of the proposed algorithm.

**Index Terms**—System Reliability, Probability Propagation, Network Graphs, Message Passing, Simulation.

## I. INTRODUCTION

Infrastructure systems are becoming more complex with technological advancements and stricter regulatory policies, there is a need for efficient and adaptable methods to assess the reliability of these systems. In this work, we consider the reliability of a system in terms of the probability of a system able of performing all tasks related to its intended operational function within specified system conditions and time frames. This view of reliability allows for the understanding of critical components, assessing redundancy strategies as well as providing valuable insights into other system performance measures.

Reliability can broadly be classified into two categories : analytical and simulation-based strategies [1]–[3]. Simulation methods such as the Monte-Carlo-based approach [4] aim to solve for system reliability by running a number of simulated random processes to numerically calculate an approximation for system reliability. On the other hand, analytical methods such as fault tree analysis [5] and reliability block diagrams [6] leverage on mathematical formulations and logic diagrams to assess the reliability of the system. Hybrid methods for system reliability exist such as methods based on the concept of survival signature [7], [8].

Directed acyclic graphs (DAG) are graphical structures that can be used to model these types of systems. DAGs are a type of graph structure consisting of nodes connected by directed edges where the acyclic nature of the directed edges ensures that when traversing forward in the graph from a node  $v$ , there is no sequence of directed edge that leads back to  $v$ . DAGs are commonly used to represent complex systems where nodes can represent components, elements, or sub-processes and the directed edges can represent dependencies and relationships encoded between these elements. In DAGs, the flow of the graph is from source nodes to sink nodes. Source nodes are defined as nodes with no incoming directed edges and sink nodes refer to nodes with no outgoing directed edges. In the context of DAGs, we take each component or tasks in the system as nodes in the graph and the edges of the DAG represents the relationship and dependencies between nodes.

In this paper, we present an analytical approach called Information Propagation Method (IPM) to evaluating system reliability in systems represented by Directed acyclic graphs (DAGs). In IPM, we adopt the message-passing concept from the belief propagation algorithm [9]–[12]: messages are passed based on a topological sorting of the nodes contained in the graph. The goal of the IPM algorithm is to efficiently propagate these probabilities through the network to calculate and update the individual reliability of each node in the network up to and including the sink nodes. We consider the reliability of individual components in a DAG to be the probability of a node's success based on its initial probability and the incoming dependencies from its neighbouring nodes. Thus, we take the reliability of the system to be the probability of success of all tasks to be carried out within the system where the sink nodes represent the final tasks. The primary advantage of the IPM is its computational efficiency making it adaptable for real-time analysis of large and complex systems. In this algorithm, a rule for prioritizing a specific set of nodes is enforced to ensure efficient and optimal processing of nodes.

## II. THE INFORMATION PROPAGATION METHOD

IPM operates by iteratively computing local solutions within the graph at a sub-graph level to exploit node to node relationships and dependencies. The local solution is propagated as

messages throughout the graph until a global system solution is reached. This makes the algorithm useful for the investigation of the represented system and assessment of key metrics such as system reliability. The “information” here is related and derived from the system problem to be solved and specific system attributes to be assessed. During each iteration of the algorithm, the sets of nodes that are processed are determined strictly based on the so-called propagation state of the node. This restriction enforces a specific belief-updating order which minimizes the number of times nodes are processed thereby optimizing the iteration process overall. Beyond reasoning about the whole system, IPM also considers individual nodes and sub-systems within the graph as the belief of every node in the graph is updated. Although the metric and information being propagated in this paper is the reliability of the nodes and edges, IPM can be applied to various other types of information depending on the network and metrics to be investigated.

#### A. Topological Sorting in Directed Acyclic Graphs

Here, we prove by induction that for a given DAG, there exists at least one topological sorting of nodes. In this ordering, we prioritise a specific dependency-aware ordering of traversal: if a node  $v$  is a successor to another node  $u$  (i.e., there is a directed edge from  $u$  to  $v$ ), then  $u$  must be traversed and must appear before  $v$  in the ordering. Let  $G$  be a non-empty DAG with  $n$  nodes. In the case of only one node, the sorting contains only the node itself. *Inductive step:* Assuming that every DAG with  $n-1$  nodes has at least one valid sorting, then also a DAG with  $n$  nodes has a valid sorting.

*Proof.* Since  $G$  is a DAG, then there must exist at least one node with no incoming edges. We refer to this node as  $ns$ . If we removed  $ns$  and all its outgoing edges from  $G$ , we are left with a new graph  $G'$ . Due to the properties of DAGs, the resulting graph  $G'$  with  $n-1$  nodes is also a DAG.  $\square$

Since  $G'$  is also a DAG, we may use sorting approaches such as Khan’s algorithm [13] to get a valid topological sorting  $T'$  of  $G'$ . Finally, adding the node  $ns$  to the beginning of  $T'$  results in a valid topological sorting for the original DAG. By design, the node  $ns$  has no incoming edges and therefore it meets the condition that every node in the sorting must only be added to the sorting after all of its dependent nodes have been added to the sorting. Thus, proving that if there exists a valid sort of  $G'$  with  $n-1$  nodes then there also exists a valid sort for  $G$  with  $n$  nodes. The propagation states in IPM adopt this “dependency-aware topological sorting,” where a node is only updated and traversed when all its dependencies (incoming edges) are satisfied. This sorting ensures that all dependencies of a node are considered in reduced time, as nodes are processed the minimum number of times possible depending on the complexity of the DAG structure.

#### B. Propagation States and Iteration Process

The propagation iterations begin with the source nodes, and converges at the sink nodes when all other nodes in the graph

have been updated. During each iteration, nodes sharing a directed edge are referred to as parent to child node pairs. The parent node represents the node at the base of the directed edge that points to the child node and the flow of traversal in the graph is always from parent to child. The iterative process in the algorithm involves simultaneously updating the belief of a set of nodes and propagating their updated messages to its children. This belief of each node is derived from the node’s initial condition and incoming messages from all of its parents. With the topological sorting described above, at the beginning of every iteration, a node is in one of three states: *ready*, *not ready* and *satisfied*. Only *ready* nodes can be processed, updated and propagated from. A node is *ready* when the belief of all its neighbouring nodes with incoming edges (parents) have been satisfied and propagated. After a node’s belief has been updated and propagated to its children, it becomes a *satisfied* node during the current iteration and no longer needs to be considered for future iterations.

At the beginning of the algorithm, every node but the source nodes are in the *not ready* state as source nodes do not have any parents or incoming edges. Therefore, the iteration starts at the sources and propagates towards their respective child nodes and repeats until convergence when all sink nodes have been updated. During each iteration the only nodes that are considered and added to the working graphs are the *ready* nodes whose parents have been previously satisfied. Since a node can only be *ready* and then updated when its own parents are satisfied, all dependencies related to a node are always handled before the node’s belief is updated. An example of the updating process is shown in Fig. 1 where the working graph consists of satisfied nodes from previous iterative steps and their respective outgoing edges.

#### C. Decomposition of DAGs: Tree Structures

In graph theory, a tree is a type of graph that is both acyclic and connected (i.e., there exists a path between every pair of nodes). In IPM, we extend the definition of tree structures to refer to a DAG that retains the hierarchical structure of tree graphs whilst still allowing for multiple source and sink nodes within the graph. In these structures, there is exactly one path between any pair of non-sink nodes and therefore a clear flow of information from sources to sinks. The IPM algorithm is guaranteed to converge to an exact solution for DAG systems that are “tree-like” in nature. More specifically, the algorithm’s solution is exact if a DAG satisfies one of the following conditions:

- The graph has a tree-like structure where there is only one unique path between any two nodes e.g., Fig. 2 (left)
- If the sink and/or source nodes are excluded from the graph and the resulting DAG has a tree like structure i.e semi tree-like as shown in Fig. 2 (right)

#### D. Complex subgraphs in DAGs

The complexity of non-tree structured DAGs is an effect of the presence of complex subgraph types in the network. In this work, we focus on a specific type of subgraph that we refer

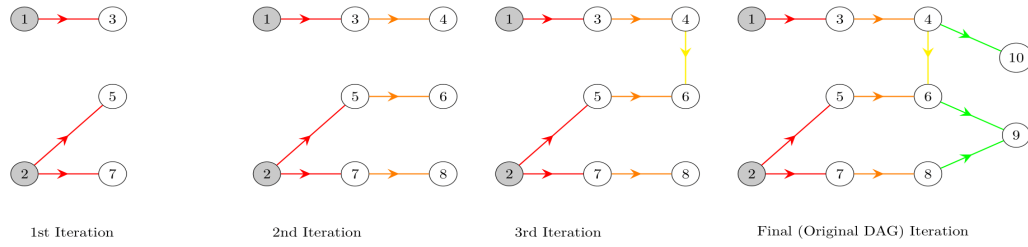


Fig. 1. Working graphs during iteration process of a DAG system.

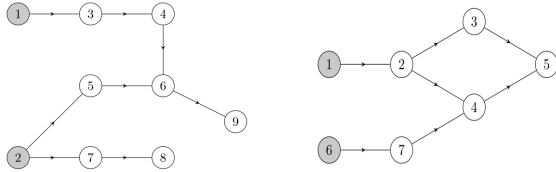


Fig. 2. A Tree-like (on the left) and a Non-tree DAG structure (on the right).

to as “diamonds”. In non-tree DAGs, the complexity arises mainly from *fork* and *join* nodes that have multiple incoming and/or outgoing edges. A node is a *fork* if it has more than one outgoing edge (i.e., it has more than one child node). A *join* node refers to a node with multiple incoming edges. A node can simultaneously be both a *fork* and a *join*. We define a *diamond* subgraph  $D$  a part of a graph  $D$  that begins with a *fork* node and ends with a *join* node both of which are connected by at least two concurrent independent paths that diverge at the *fork* and converge back at the *join* node. The diamond shape refers to the way the subgraph is visualized when drawn with respect to the topological sorting (Fig. 3). In graph theory terms, the subgraph can be thought of as a *divergent-convergent subgraph* where every other node shares the fork node as a common ancestor and the join as a common descendant. Diamonds can be chained, nested, induced or interconnected. An induced diamond is one where all the non-fork nodes in the diamond do not depend on other nodes that are not part of diamond node set. Chained diamonds are types of induced diamonds where two diamonds share only one node in common, in this case, the join node of the first diamond is also the fork of the second. In nested diamonds, all nodes and edges of a smaller induced diamond are contained within a larger diamond. Inversely, more complex diamond types can be interconnected where multiple diamond subgraphs share nodes or edges with a different diamond as shown in Fig. 3.

### E. Handling Diamond Subgraphs

In complex DAGs where diamond structures exist, the IPM solves the subgraph by isolating the diamond and treating this subgraph as its own DAG subsystem: the *fork* of the diamond becomes a source node and the *join* of the diamond become

the sink. A recursive procedure solves for the sink/join node of the diamond. The updating procedure in a non-tree DAG is as follows:

- If a ready node is a *join* node, mark as potential for diamond *sink*.
- For each marked node, check the list of *satisfied* nodes for any *fork* nodes that forms a *fork-join* pair with the potential join node.
- If any *fork-join* pair is identified, check for all unique independent paths that start and end at these pair of nodes.
- If more than one path is found, check the combinations of *fork-join* paths to verify if they form a diamond subgraph.
- For each diamond identified, mark the set of nodes that form the diamond subgraph.
- Solve the identified diamond subgraph by the IPM algorithm.
- After all the diamonds have been processed for the ready join nodes, the iteration process continues until the main algorithm converges and original graph connectivity is restored.

When diamonds are chained or interconnected, the rules regarding propagation states still apply and dependencies must always be satisfied first. For example, in the third graph with nested diamonds in Fig. 3, the diamond subgraph formed by the nodes 2, 4, 5 and 6 is nested within the larger diamond structure, therefore the join node of the first diamond (node 6) will always be handled first before the join (node 8) of the larger diamond can be processed. Similarly, for diamonds that share edges or nodes, the propagation state of the nodes is respected both within the local scope of diamond as well as the global scope within the main DAG graph. This process of handling diamonds by isolating and resolving the subgraph as a subsystem is comparable to graph decomposition concepts discussed in [14]–[16] where the diamond can be thought of as a meta-node that encompasses the parallel paths that form the diamond and thereby simplifying the graph structures.

### III. EVALUATING RELIABILITY IN DAG NETWORKS

In this paper, we take a node’s reliability to be the updated probability of success based on the node’s prior state and the incoming messages from its dependencies. Nodes in the graph can represent components of the system and the directed edges represent operational dependencies and the flow of

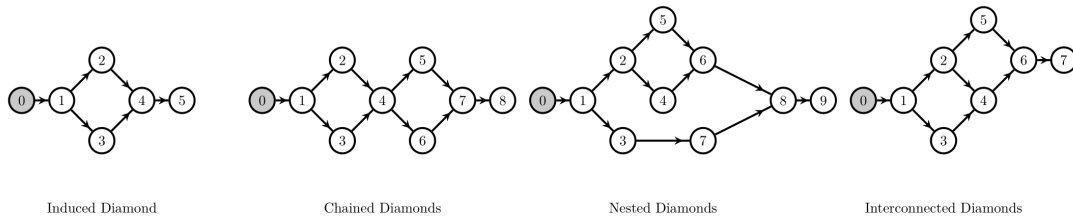


Fig. 3. Different types of diamonds possible in a complex DAG

information in the network. In such interpretations, source nodes often symbolize components of the system that initiate or generate the system procedure while the sink nodes are components or tasks which terminate or consume the end-product or service. The reliability of these types of systems can be defined in terms of the propagated probability from source to sink nodes. In addition to the initial conditions of the source nodes, other nodes are typically associated with operation probabilities which often depends on factors such as component age, quality, and other environmental conditions. Although the scope of this paper focuses on applying the IPM approach for assessing system reliability, the algorithm itself is adaptable to different information metrics such as availability in systems that can be modelled as DAGs.

#### A. Evaluating reliability with IPM

For propagating system reliability with the information propagation method, we must first derive definitions and calculations for updating the belief and messages. We adopt the series-parallel reliability calculation with reliability block diagrams in [17], [18] where series reliability is taken as a product of all individual reliabilities. In parallel system configurations, failed components can be compensated for by redundant components. Since a system in a parallel configuration will only fail if all paths from source nodes fail, we can derive reliability from the system's failure probability (also known as unreliability). Therefore, we can infer the system reliability from the complement of the unreliability since the sum of the reliability and failure probability are mutually exclusive events. The reliability messages passed between nodes is defined as:

$$\text{Message}_{p \rightarrow c} = \text{Belief}_p \times \text{Edge}_{p \rightarrow c} \quad (1)$$

and

$$\text{Belief}_n = \text{Prior}_n \times \left( 1 - \prod_{i=1}^n (1 - \text{Message}_{i \rightarrow n}) \right) \quad (2)$$

where the  $\text{Message}_{p \rightarrow c}$  is message from a parent node  $p$  to a child node  $c$ ,  $\text{Belief}_p$  is the updated belief of the parent node  $p$  and  $\text{Edge}_{p \rightarrow c}$  is the edge probability between the parent node  $p$ , the child node  $c$ ,  $\text{Belief}_n$  is the updated belief of node  $n$  and finally  $\text{Prior}_n$  is the initial reliability of node  $n$ .

#### B. Propagating reliability in diamond subgraphs

The reliability of the subsystem is considered in terms of the probability of success of the path between the diamond's *fork* and its *join* node. To this end, during the iteration step where a diamond's sink needs to be updated, the diamond is still considered as its own subsystem where the fork and join are the source and sink respectively. The algorithm is then applied to the subsystem to solve for the belief of the join node denoted as

$$B_{\text{join}} = S_{\text{diamond}} \times B_{\text{fork}} + F_{\text{diamond}} \times (1 - B_{\text{fork}}) \quad (3)$$

where  $B_{\text{join}}$  represents the updated belief or reliability of the join node,  $S_{\text{diamond}}$  is the belief at the join node when the fork node is fully operational,  $B_{\text{fork}}$  is the actual updated belief or reliability of the fork node and  $F_{\text{diamond}}$  is the belief at the join node when the fork node's is in a non-operational state.

For calculating reliability, the algorithm is applied to the subsystem under two distinct scenarios. The first scenario  $F_{\text{diamond}}$ , considers the fork's initial reliability and probability of success to be zero, signifying a failure state of the subsystem's source node. The second scenario  $S_{\text{diamond}}$ , assumes the fork of the diamond has a success probability of one, indicating a guaranteed operational states for the node. These two scenarios encapsulate the extreme cases in the reliability analysis and thus provide bounds within which the actual join node probability is expected to lie. The values are then combined to calculate the reliability (belief) at the join node as shown in Eq. (3). It should be noted that when a diamond is induced, then the failure belief will always be equal to zero and the IPM does not need to be applied twice since the probability of success of the node is exclusively dependent on the paths contained within the diamond.

#### C. Evaluating reliability in a complex DAG

The system in Fig. 4 represents a network of natural gas facilities adapted from [19]. The gas network consists of 16 nodes that symbolize gas and regulator stations, which are supplied by the source node  $S$  and these nodes are connected by transmission lines. We further simplify the DAG by treating the transmission line nodes as edge probabilities. We assume an initial probability of 1 for all nodes and a consistent edge probability of 0.9. Table I shows the calculated reliability of every node in the graph based on the propagated probabilities of its dependencies obtained by the Monte Carlo method and with the IPM algorithm.

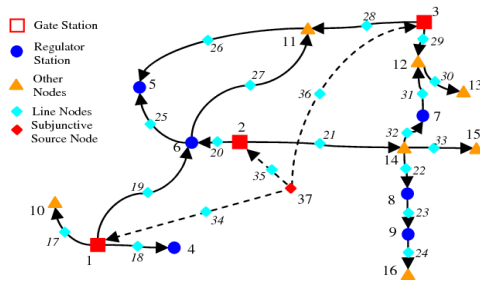


Fig. 4. DAG of a natural gas network.

TABLE I  
RELIABILITY ANALYSIS OF THE NATURAL GAS NETWORK

| Node | IPM                   | Monte Carlo                 | Monte Carlo                |
|------|-----------------------|-----------------------------|----------------------------|
|      | Time = 19.500 $\mu$ s | N=10,000<br>Time= 95.895 ms | N=100,000<br>Time= 1.015 s |
| 1    | 0.9                   | 0.8981                      | 0.90102                    |
| 2    | 0.9                   | 0.8968                      | 0.89958                    |
| 3    | 0.9                   | 0.9066                      | 0.90028                    |
| 4    | 0.81                  | 0.8122                      | 0.81089                    |
| 5    | 0.978929631           | 0.9771                      | 0.97858                    |
| 6    | 0.9639                | 0.9622                      | 0.96397                    |
| 7    | 0.7290000000000001    | 0.7254                      | 0.72843                    |
| 8    | 0.7290000000000001    | 0.7277                      | 0.72869                    |
| 9    | 0.6561000000000001    | 0.6581                      | 0.65558                    |
| 10   | 0.81                  | 0.8075                      | 0.81087                    |
| 11   | 0.9748269             | 0.9766                      | 0.97456                    |
| 12   | 0.934659              | 0.935                       | 0.93334                    |
| 13   | 0.8411931             | 0.8397                      | 0.83997                    |
| 14   | 0.81                  | 0.8065                      | 0.81117                    |
| 15   | 0.7290000000000001    | 0.7293                      | 0.73006                    |
| 16   | 0.5904900000000002    | 0.5918                      | 0.59092                    |
| 17   | 1.0                   | 0.0001                      | $1.0 \times 10^5$          |

#### IV. CONCLUSION AND FUTURE WORK

In this paper, we propose the Information Propagation Method for efficient reliability analysis on directed acyclic graphs. We take the reliability of a DAG system to be related to the probability of a successful path from the source nodes to the sinks to symbolise the successful operation of system functions as intended. The primary advantage of the algorithm is its computational time, which is directly related to the number of nodes and diamond subgraphs in the target network. IPM optimises the belief-updating process of each node by restricting the set of nodes that can be updated in any one iteration, thereby minimizing the number of times a node has to be processed overall. Future work might involve extending the algorithm's capabilities to automatically detect and identify nodes in the graph that constitute a diamond subgraph during the iteration process. We also plan to broaden the type of information represented in nodes and edges within the network, moving beyond just discrete values.

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#### REFERENCES

- [1] P. K. Das and W. Zhang, "Guidance on Structural Reliability Analysis of Marine Structures," Department of Naval Architecture & Marine Engineering, Universities of Glasgow and Strathclyde, October 2003. Available: <https://api.semanticscholar.org/CorpusID:174772228>.
- [2] S. H. Hoseinie, H. Al-Chalabi, and B. Ghodrati, "Comparison between Simulation and Analytical Methods in Reliability Data Analysis: A Case Study on Face Drilling Rigs," Data, vol. 3, no. 2, Article no. 12, 2018. Available: <https://www.mdpi.com/2306-5729/3/2/12>. DOI: 10.3390/data3020012.
- [3] A. Ostfeld, D. Kogan, and U. Shamir, "Reliability simulation of water distribution systems – single and multiquality," Urban Water, vol. 4, no. 1, pp. 53-61, 2002. DOI: 10.1016/S1462-0758(01)00055-3.
- [4] A. J. Sobey, J. I. R. Blake, and R. A. Shenoi, "Monte Carlo reliability analysis of tophat stiffened composite plate structures under out of plane loading," Reliability Engineering & System Safety, vol. 110, pp. 41-49, 2013. DOI: 10.1016/j.res.2012.08.011.
- [5] F. P. García Márquez, I. Segovia Ramírez, B. Mohammadi-Ivatloo, and A. Pliego Marugán, "Reliability Dynamic Analysis by Fault Trees and Binary Decision Diagrams," Information, vol. 11, no. 6, Article 324, 2020. DOI: 10.3390/info11060324.
- [6] V. de Vasconcelos, W. A. Soares, A. C. Lopes da Costa, and A. L. Raso, "Use of Reliability Block Diagram and Fault Tree Techniques in Reliability Analysis of Emergency Diesel Generators of Nuclear Power Plants," International Journal of Mathematical, Engineering and Management Sciences, 2019. Available: DOI: 10.33889/IJMEMS.2019.4.4-064.
- [7] H. George-Williams, G. Feng, F. Coolen, M. Beer, and E. Patelli, "Extending The Survival Signature Paradigm To Complex Systems With Non-repairable Dependent Failures", Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability, vol. 233, no. 4, pp. 505-519, Aug. 2019, DOI: 10.1177/1748006X18808085.
- [8] E. Patelli, G. Feng, F. Coolen, and T. Coolen-Maturi, "Simulation Methods for System Reliability Using Survival Signature", Reliability Engineering & System Safety, 2017, DOI: 10.1016/j.res.2017.06.018.
- [9] J. S. Yedidia, W. T. Freeman, and Y. Weiss, "Understanding Belief Propagation and Its Generalizations," in Exploring Artificial Intelligence in the New Millennium, San Francisco, CA, USA: Morgan Kaufmann Publishers Inc., 2003, pp. 239-269.
- [10] M. Steinder and A.S. Sethi, "Probabilistic fault localization in communication & systems using belief networks," IEEE ACM Transactions on Networking, vol. 12, no. 5, pp. 809-822, 2004. DOI: 10.1109/TNET.2004.836121.
- [11] J. Pearl, "Fusion, propagation, and structuring in belief networks," Artificial Intelligence, vol. 29, no. 3, pp. 241-288, 1986. DOI: 10.1016/0004-3702(86)90072-X.
- [12] J. Coughlan, "A Tutorial Introduction to Belief Propagation," 2023.
- [13] A. B. Kahn, "Topological Sorting of Large Networks," Commun. ACM, vol. 5, no. 11, pp. 558-562, Nov. 1962. DOI: 10.1145/368996.369025.
- [14] S. Arnborg, D. G. Corneil, and A. Proskurowski, "Complexity of Finding Embeddings in a k-Tree," SIAM Journal on Algebraic Discrete Methods, vol. 8, no. 2, pp. 277-284, 1987. DOI: 10.1137/0608024.
- [15] D. J. Rose, "Triangulated graphs and the elimination process," Journal of Mathematical Analysis and Applications, vol. 32, no. 3, pp. 597-609, 1970. DOI: 10.1016/0022-247X(70)90282-9.
- [16] B. Alspach and H. Gavlas, "Cycle Decompositions of  $K_n$  and  $K_n - I$ ," Journal of Combinatorial Theory, Series B, vol. 81, no. 1, pp. 77-99, 2001. DOI: <https://doi.org/10.1006/jctb.2000.1996>.
- [17] J.P. Signoret and A. Leroy, "Reliability Block Diagrams (RBDs)," in Reliability Assessment of Safety and Production Systems: Analysis, Modelling, Calculations and Case Studies, Springer International Publishing, Cham, 2021, pp. 195-208. DOI: 10.1007/978-3-030-64708-7-15.
- [18] E. G. Frankel, "Reliability of Series and Parallel Systems," in Systems Reliability and Risk Analysis, Springer Netherlands, Dordrecht, 1984, pp. 73-106. DOI: 10.1007/978-94-009-6920-9-5.
- [19] L. Chang and J. Song, "Matrix-based system reliability analysis of urban infrastructure networks: A case study of MLGW natural gas network," 2007. [Online]. <https://www.ideals.illinois.edu/handle/2142/5123>.
- [20] Y. Tong and I. Tien, "Probability Propagation Method for Reliability Assessment of Acyclic Directed Networks," ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering, vol. 5, no. 3, p. 04019011, Sep. 2019. DOI: 10.1061/AJRUA6.0001017.